



KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

Enriching Lives

Description : Differential Pressure (DP) Switch Selection and Set Points addition		
Document No : ESP-20-21-008	Revision : R00	Date : 29/08/2020
Prepared By : SSM	Approved By: SNM	Sheet : 1 of 2

Engineering Specification

Differential pressure switches for evaporator and condenser ensure that there is no insufficient water flow through both heat exchangers.

Procedure to select DP switches for Evaporator and Condenser Flow

Constant Evaporator and Condenser Flow

- Calculate the Pressure drop at 90% flow of the design flow stated in the Technical Data Sheet or Software Output.
- Calculate the minimum allowable flow using HTC/Software.
- Calculate the Pressure drop corresponding to the minimum allowable flow.
- If the difference between the pressure drop values calculated at 90% Flow and Minimum allowable flow is greater than or equal to 0.1bar, then select the Regular DP switch.
- If the difference between the pressure drop values calculated at 90% Flow and Minimum allowable flow is less than 0.1bar, then select the Low range DP switch.

Primary Variable Flow for the Evaporator

- Calculate the pressure drop values at the Minimum variable flow.
- Calculate the Minimum allowable flow from HTC/Software and its corresponding pressure-drop value.
- If the difference between the pressure drop values calculated at Minimum variable flow and Minimum allowable flow is greater than or equal to 0.1 bar, then select the Regular DP switch.
- If the difference between the pressure drop values calculated at Minimum variable flow and Minimum allowable flow is less than 0.1 bar, then select the Low range DP switch.

Note: Please discuss if the difference between pressure drop values is less than 0.05 Bar.



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Procedure to Add DP Switch Set Points

Constant Evaporator and Condenser Flow

- If a Regular DP switch is provided, then reduce 0.1 bar from the pressure drop value calculated at 90% of the design flow to get the DP switch set point.
- If Low range DP switch is provided, then reduce 0.04 bar from the pressure drop value calculated at 90% of the design flow to get the DP switch set point.

Primary Variable Flow for the Evaporator

- If a Regular DP switch is provided, then reduce 0.05 bar from the pressure drop value calculated at the Minimum variable flow to get the DP switch set point.
- If Low range DP switch is provided, then reduce 0.025 bar from the pressure drop value calculated at the Minimum variable flow to get the DP switch set point.